

# Technology #165: apply a magnetorheological (MR) fluid coating to the rail surface and energize it with a magnetic field during service to generate a compressive stress field at potential crack tips, thereby inducing a crack tip shielding effect under cyclic loading

**Technology Domain:** mechanical\_structural

**Mechanism:** PROPOSED architecture (no physical system exists): load path / structural backbone; actuation or compliance element; guidance / bearing interfaces; fatigue-critical features register — realizing: apply a magnetorheological (MR) fluid coating to the rail surface and energize it with a magnetic field during service to generate a compressive stress field at potential crack tips, thereby inducing a crack tip shielding effect under cyclic loading

**Maturity:** ENGINEERING\_DEFINITION

**Current State:** No physical prototype. Computational model and engineering definition exist.

**Key Risk:** killer experiment fails: falsification\_test\_from\_candidate (decision scores: EIG 0.44573539382492045, EIG-per-cost 0.44573539382492045; the falsification test carries the pre-registered pass/fail before any transfer)

**Next Experiment:** first physical article implementing the proposed intervention (GEOMETRY ABSENT — must be designed)

**Buyer Receives:** Buyer receives: canonical invention specification (with epistemic classes).; Buyer receives: engineering specification (this document, all blocks classed).; Buyer receives: design input/output graph with explicit IDs.; Buyer receives: engineering reasoning chains (claim->principle->model->input->assumption->output->failure->verification).; Buyer receives: killer-experiment definition and its EIG basis.; Buyer receives: evidence custody references and hashes.; Buyer receives: unresolved-uncertainty register.; Buyer receives: domain equation set (symbolic; sources and applicability judgments identified).

**Buyer Must Build:** Buyer must create: actual geometry and detailed design.; Buyer must create: sourced engineering parameters and acceptance thresholds.; Buyer must create: physical prototypes.; Buyer must create: all verification and validation results.; Buyer must create: manufacturing process qualification.; Buyer must create: regulatory pathway determination and submissions.

**Transfer Posture:** Sponsored Validation

**Target Buyer:** domain-appropriate technology buyer (to be identified by portfolio owner)

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